

Assessment of the Protective Potential of *Premna tomentosa* (L. Verbenaceae) Extract on Lipid Profile and Lipid-Metabolizing Enzymes in Acetaminophen-Intoxicated Rats

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ABSTRACT

Objectives: The liver is often damaged by environmental toxins, poor eating habits, alcohol and over-the-counter drug use that damage and weaken the liver, leading to important public health problems such as hepatitis, cirrhosis, and alcoholic liver diseases. It is cardinal to treat liver disorders, because it affects the biochemistry of the cell directly. Damage to the liver can be prevented by including a balanced diet that includes nutrients and herbs that support a healthy liver. *Premna tomentosa* (PT) is one such herbal drug used widely in India for the treatment of liver disorders, and we have already reported the hepatoprotective potential and antioxidant property of methanolic extract of PT leaves. Because injury to the liver can promote a variety of reactions with consequent effect on lipids, the present study was designed to elucidate the hypolipidemic effect of PT extract in acetaminophen (AA)-induced hepatotoxicity in rats.

Design and subjects: Animals were pretreated with PT extract (750 mg/kg, orally) for 15 days and then induced with hepatotoxicity by AA (640 mg/kg, intraperitoneally).

Results: PT extract pretreatment significantly inhibited induced alterations in the levels of cholesterol, triglycerides, free fatty acids, phospholipids, serum lipoproteins, and lipid-metabolizing enzymes.

Conclusions: The results indicate that PT extract improves lipid metabolism and has the potential for use in hepatic disorders.

INTRODUCTION

The liver plays an astonishing array of vital functions in sustenance of the body. The liver is often affected by a multitude of environmental pollutants, drugs, et cetera, all of which place a burden on this vital organ that can damage and weaken the liver and eventually lead to hepatitis or cirrhosis (Zimmerman and Ishak, 1987). Although a number of strategies are envisioned to protect the liver against these agents, treatment options are often problematic, because of the profound side-effects of the chemicals used. In recent years, researchers have examined the effects of plants

traditionally used by indigenous healers and herbalist, because protecting against these attacks can be accomplished by consumption of a balanced diet that includes nutrients and herbs that support a healthy liver (Treadway, 1998).

Premna tomentosa (PT) (L. Verbenaceae), commonly called *Krishnapalai* and *Pudangainari* (it is available easily in the wild), is used extensively in Indian Siddha medicine for the treatment of various disorders. The leaves of PT—both as a monopreparation and as *Pudangainari kudineer* (a polyherbal preparation) are given for treating liver and spleen enlargements (Mudaliar, 1998). The leaves also possess diuretic and vulnerary properties (Shanmugavelu,

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1987). An extract of the inner bark is stated to arrest diarrhea and a decoction of the roots is given to treat stomachache (Wealth of India, 1989). Phytochemical screening of the leaves revealed the presence of D- and DL-limonene (57.8%), β -caryophyllene (17.2%), a cadalene-type sesquiterpene (7.8%), sesquiterpene tertiary alcohol (5.6%), and aditerpene (5.5%) (Lakshminarayan and Muthana, 1953). Earlier studies have shown that the methanolic extract of PT leaves afforded protection against acetaminophen (AA)-induced hepatotoxicity in rats (Devi and Devaki, 1998), which might be the result of its antioxidant property (Devi et al., 1998). The extract also possesses immunomodulatory effects in splenic lymphocytes (Devi et al., 2003a) and antinociceptive and hypnotic effects (Devi et al., 2003b). Because the liver plays an important role in lipid metabolism, the effect of methanolic extract of PT leaves on balancing the levels of lipids were assessed in AA-intoxicated rats.

MATERIALS AND METHODS

Animals

Male Albino Wistar rats weighing 100–150 g were obtained from Fredrick Institute for Plant Protection and Toxicology, Padappai, Chennai, India. They were acclimated to laboratory conditions for 10 days during which they were fed commercial pelleted diet (M/s Hindustan Foods Limited, Bangalore, India) and water *ad libitum*.

Preparation of plant extract

PT leaves were collected from a rural area near Palayamkottai, Tamil Nadu, India and authenticated by Dr. Usman Ali, Drug Research Scheme, Multidisciplinary Unit, Central Research Institute for Siddha, Chennai, India, where a voucher specimen of the plant has been deposited. The leaves were shade-dried and the powdered leaves were extracted with 100% methanol to a syrupy mass. The last traces of the solvent were removed in vacuum with an approximate yield of 12.4%.

Study design

The animals were divided into four groups comprising six animals in each group. Group 1 (vehicle control) received olive oil and the vehicle, 50% propylene glycol, orally. Group 2 was given olive oil (7.5 mL/kg) for 15 days and AA (640 mg/kg of body weight, orally) administered 24 hours post-treatment of the last dose. Group 3 was treated similarly to group 2 except that the plant extract (750 mg/kg of body weight, orally) (Devi and Devaki, 1998) was administered instead of olive oil. Group 4 was given plant extract alone for 15 days and served as drug control group. Animals were sacrificed by cervical decapitation, 24 hours

after the last dosage of drug or vehicle administration. The serum and liver (homogenized in 0.1 mol. Tris, pH adjusted to 7.4 with HCl) obtained from the animals were used for the estimation of the following parameters.

Estimation of lipids

Total lipid content (Folch et al., 1957), cholesterol (Mookherjee and Sadhu, 1955), triglycerides (Foster and Dunn, 1973), free fatty acids (Hron and Menahan, 1981), and phospholipids (Zilversmit and Davis, 1950) were estimated in liver homogenate and serum.

Estimation of lipid metabolizing enzymes

Lecithin:cholesterol acyl transferase (LCAT) and hepatic triglyceride lipase (HTGL) were assayed in liver using the method of Hitz et al. (1983) and Schmidt et al. (1974), respectively.

Estimation of lipoproteins

High-density lipoprotein (HDL) in serum was estimated by using Human Kit (Mannheim, Germany), and the levels of very low density lipoprotein (VLDL) and low-density lipoprotein (LDL) were calculated from that following manufacturer's instructions.

Statistical analysis

The results are expressed as mean $\pm$ standard deviation (SD) and all the statistical comparisons were made by means of Student's *t* test.

RESULTS

Effect of the extract on lipid metabolizing enzymes

AA intoxication tended to decrease the activities of LCAT (vehicle control 115.23 ± 10.7 , AA 85.62 ± 8.2) and HTGL (group 1 14.82 ± 1.1 , AA 9.83 ± 0.87) significantly ($p < 0.001$) in group 2 rats (Fig. 1), whereas the activities were comparable to the control group in the rats pretreated with PT extract.

Effect of the extract on the levels of lipids

Acute administration of AA caused a significant increase in the levels of total lipids ($p < 0.001$), cholesterol ($p < 0.001$), triglycerides ($p < 0.001$), free fatty acids ($p < 0.001$) and phospholipids ($p < 0.01$) in liver compared to group 1 vehicle-treated controls (Figs. 2 and 3). In serum, the same changes were observed in the levels of cholesterol, triglycerides, free fatty acids and phospholipids, except for the decrease in the levels of ester cholesterol ($p < 0.001$)

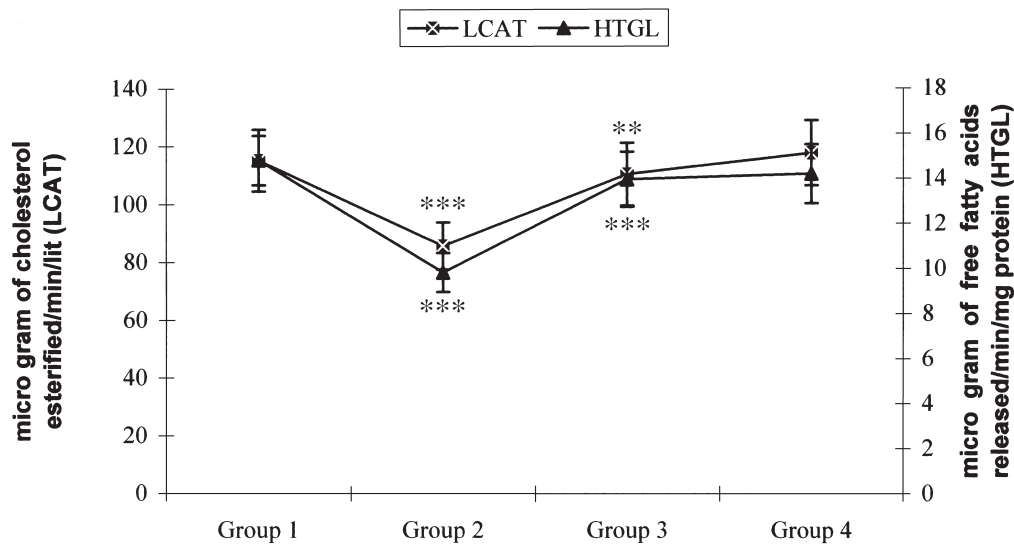


FIG. 1. Activities of serum lecithin:cholesterol acyl transferase (LCAT) and hepatic triglyceride lipase (HTGL) in control and experimental group of rats. Values are expressed as Mean $\pm$ standard deviation ($n = 6$). Statistical comparisons: *** $p < 0.001$ for group 2 versus group 1 (both LCAT and HTGL) and group 3 versus group 2 (HTGL); ** $p < 0.01$ for group 3 versus group 2 (LCAT).

and phospholipids, ($p < 0.001$) in AA-treated rats when compared to group 1 control animals (Fig. 4). However PT pretreatment in group 3 rats caused a significant reversal in the levels of these lipids to near normal.

1.72) significantly ($p < 0.001$) in rats intoxicated with AA compared to the vehicle control group. By contrast, the levels of lipoproteins were attenuated in group 3 rats that were pretreated with PT extract (Fig. 5).

Effect of the extract on lipoproteins

There was an increase in the levels of VLDL (vehicle control 10.12 ± 0.96 , AA 19.26 ± 1.83), LDL (vehicle control 30.97 ± 2.85 , AA 92.65 ± 8.87) and decrease in the levels of HDL (vehicle control 31.66 ± 2.98 , AA $18.5 \pm$

DISCUSSION

The liver is critical in the normal metabolism of energy substrates, especially lipid metabolism. The serum concentration of lipids is largely a balance of synthesis and degra-

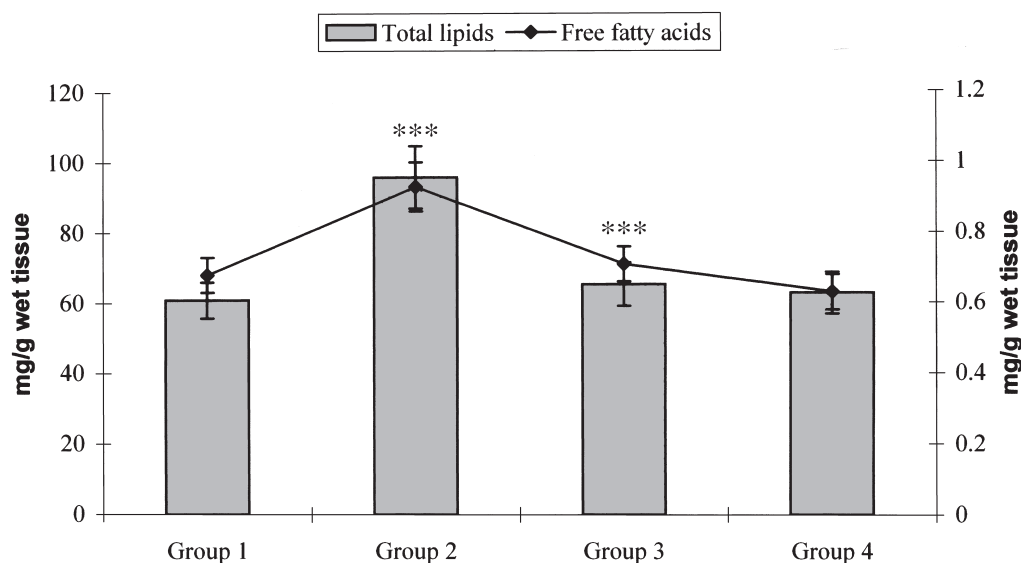


FIG. 2. Levels of total lipids and free fatty acids in the liver of control and experimental groups of rats. Values are expressed as Mean $\pm$ standard deviation ($n = 6$). Statistical comparisons: *** $p < 0.001$ for group 2 versus group 1 and group 3 versus group 2.

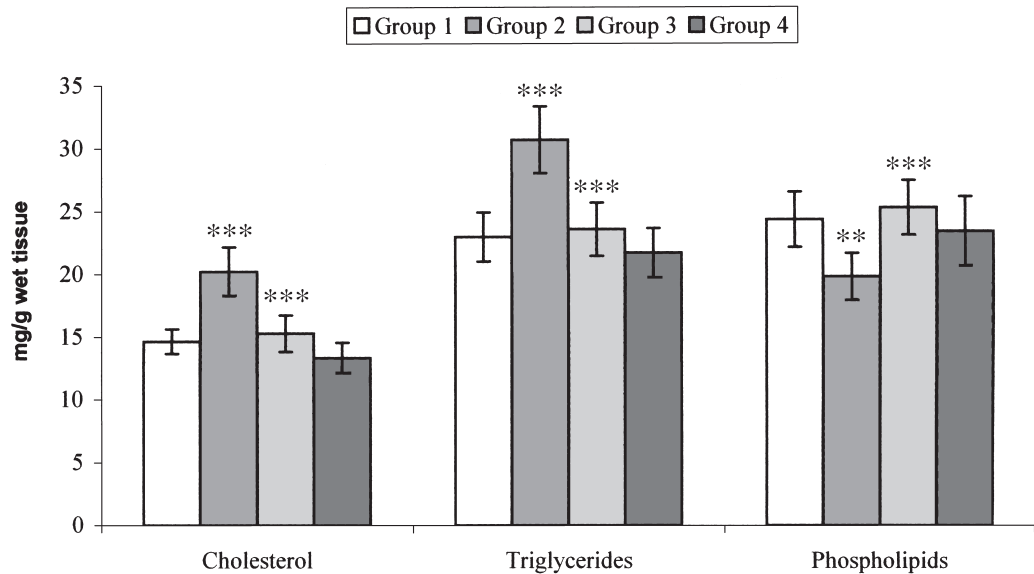


FIG. 3. Levels of cholesterol, triglycerides, and phospholipids in the livers of control and experimental groups of rats. Values are expressed as Mean $\pm$ standard deviation (n = 6). Statistical comparisons: *** p < 0.001 for group 2 versus group 1 (cholesterol and triglycerides) and group 3 versus group 2 (cholesterol, triglycerides, and phospholipids); ** p < 0.01 for group 2 versus group 1 (phospholipids).

dation by the liver. Any defect, clinical condition, or drug that alters that balance can have a profound effect on lipoprotein metabolism (Katbamna et al., 1997). Lipoproteins are metabolically diverse macromolecules that serve to transport water-insoluble lipids, mainly cholesterol ester, triglyceride, cholesterol, and phospholipid molecule in the circulation (Voet and Voet, 1995).

A key factor in maintaining the surface composition of lipoproteins is the activity of the enzyme LCAT. This enzyme is synthesized in the liver and secreted into the circulation where it catalyses the transfer of a long-chain fatty acyl group from the two-position of lecithin to the 3- β -hydroxyl group of cholesterol, forming cholesteryl ester and lysolecithin. Because hepatocytes synthesize LCAT, low

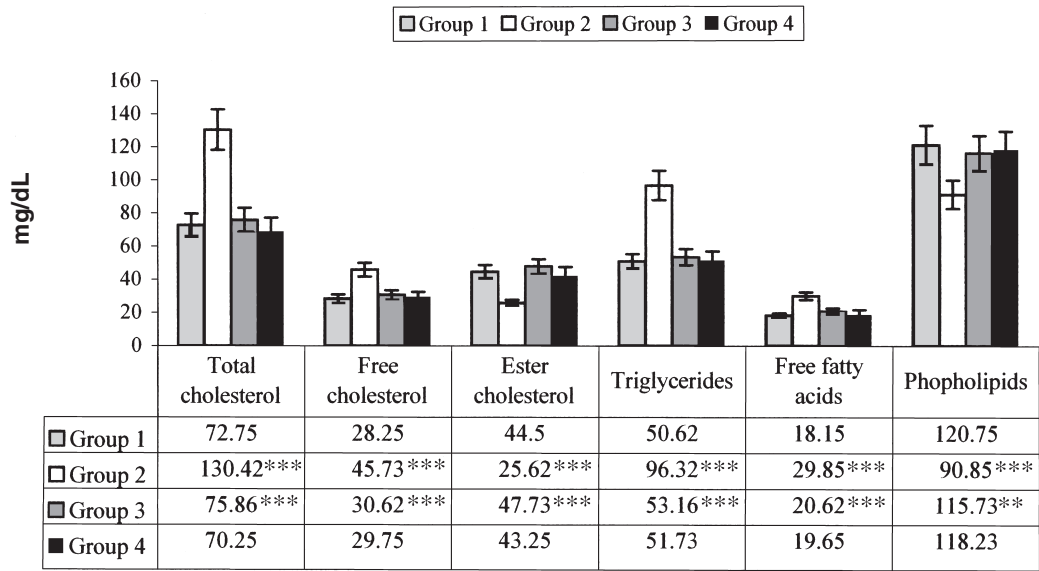


FIG. 4. Levels of cholesterol (total, free, and ester), triglycerides, free fatty acids, and phospholipids in the serum of control and experimental groups of rats. Values are expressed as Mean $\pm$ standard deviation (n = 6). Statistical comparisons: *** p < 0.001 for group 2 versus group 1 and group 3 versus group 2 (except phospholipids); ** p < 0.01 for group 3 versus group 2 (phospholipids).

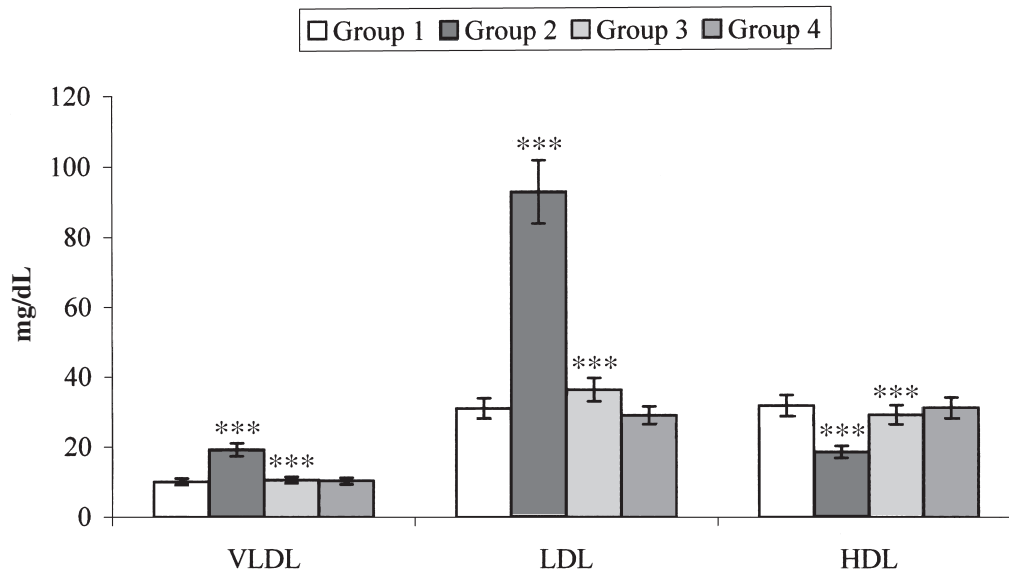


FIG. 5. Levels of lipoprotein fractions in the sera of control and experimental groups of rats. Values are expressed as mean $\pm$ standard deviation ($n = 6$). VLDL, very low-density lipoprotein; LDL, low-density lipoprotein; HDL, high-density lipoprotein. Statistical comparisons: *** $p < 0.001$ for group 2 versus group 1 and group 3 versus group 1.

LCAT activity is a common feature of acute and chronic liver diseases; indeed its decrease is a sensitive and reliable indicator of hepatic damage. Impaired hepatic synthesis leads to low amounts of LCAT protein (Owen, 1990), which might be the reason that low LCAT activity in AA-intoxicated rats. However, pretreatment with PT extract in group 3 rats increased ($p < 0.01$) the activities of LCAT that might be caused by improved synthesis of LCAT protein by the liver that is protected from damage by the extract.

As shown in Figures 3 and 4 there was an increase in the levels of cholesterol (total and free) in serum, total cholesterol in liver, and triglycerides in both serum and liver in Group 2 AA-intoxicated rats. Cholesterolemia is attributed to the failure of the liver to remove cholesterol from the blood. Increase in free cholesterol in serum is characteristic of liver diseases (McIntyre, 1978). Modest hypertriglyceridemia occurs frequently in association with hepatocellular diseases, as described in alcoholics, viral and drug-induced hepatitis (Glickman and Sebesin, 1982). In the present study, the exaggerated accumulation of triglycerides in the liver of animals treated with AA in group 2 rats may result in part from the biochemical changes that interfere with the transport of triglycerides out of the liver, whereas, the levels were brought back to near normal in group 3 rats pretreated with PT extract. The inhibitory effect of orally administered PT extract on lipid accumulation should be closely related to the preventive effect of the herbal medicine on the progression of liver injury.

A decrease in the levels of ester cholesterol was observed in serum in group 2 AA-treated rats as shown in Figure 4. Circulating cholesterol esters are formed by LCAT reaction.

LCAT deficiency is the principal explanation for decreased cholesterol esters. Thus, the decrease in LCAT activity and the parallel depression in cholesterol ester concentrations are related to the extent of liver damage (Wenegler et al., 1972). Pretreatment with PT extract in group 3 rats were able to prevent the changes induced by AA on ester cholesterol probably by the improved activities of LCAT, which was maintained by the extract.

The levels of phospholipids were decreased and free fatty acids were significantly increased ($p < 0.001$) in liver homogenate and serum in group 2 AA-intoxicated rats as shown in Figures 3 and 4 respectively. Degradation of membrane phospholipids is a two-step process involving phospholipase A₂ that removes one fatty acid followed by the removal of the second by a lysophospholipase. These enzymes have alkaline pH optima and are somewhat dependent on Ca²⁺ ions (Frei and Zahler, 1979). An increase in Ca²⁺ would activate Ca²⁺-dependent phospholipases resulting in breakdown of the cell membranes and production of free fatty acids and lysophospholipids that are toxic to cells (Buja et al., 1985). Toxic doses of AA cause total liver Ca²⁺ to accumulate by 24 hours and Ca²⁺ adenosine triphosphatase (ATPase) to increase sharply in cytosol (Ray et al., 1990), which explains the decrease in phospholipids and increase in free fatty acids in our study. Orally administered PT extract to group 3 rats caused marked increase in phospholipids with subsequent decrease in free fatty acids in serum and liver suggesting the Ca²⁺-sequestering capacity of the extract.

Plasma lipoprotein abnormalities in liver diseases change the lipid composition of erythrocyte membranes and these

changes are associated with an abnormal permeability of the membrane to sodium. There are similar changes in platelets and possibly in the membranes of other cells. Such changes could affect cellular function and be responsible at least in part, for the general cellular disturbance in liver disease (McIntyre, 1978). Because the abnormal composition of serum lipoproteins needs to be corrected, the levels of lipoproteins were estimated in experimental groups of animals.

HTGL is synthesized in the liver and binds to the luminal surface of endothelial cells in the liver sinusoids. HTGL plays a role in removing triglycerides from the partially catabolized VLDL or intermediate-density lipoprotein (IDL) and therefore plays a role in the conversion of VLDL to LDL. HTGL is also involved in metabolism of chylomicrons and VLDL remnants and also for the metabolism of HDL (Kostner, 1991). A decrease in HTGL with a subsequent increase in VLDL is noted in Group 2 AA-intoxicated rats as shown in Figures 1 and 5, respectively. The levels were brought back to normal in PT pretreated rats, which shows that the extract protects the liver cells against abnormal permeability of membranes.

In hypertriglyceridemia, VLDL concentration is elevated and a threefold increase in cholesterol ester transfer from HDL to IDL occurs, subsequently leading to enhanced formation of small dense LDL particles and lowered HDL levels (Ghatak and Asthana, 1995). HDL contains more amount of vitamin E than VLDL or LDL, and under the same experimental conditions, oxidized HDL is less cytotoxic to macrophages than the oxidized β -lipoproteins (VLDL and LDL; Hurtado et al., 1996). Furthermore, HDL acts as a scavenger of oxygen free radicals and prevents the peroxidation of β -lipoproteins (Chander and Kapoor, 1990). The enhanced levels of LDL and lowered levels of HDL in group 2 rats as shown in Figure 5 is the result of hypertriglyceridemia induced by AA intoxication. Both the increase in LDL and decrease in HDL induced by AA administration were prevented by pretreatment with PT extract in group 3 rats.

CONCLUSIONS

The protective action of PT may be caused by the prevention of free radical generation (Devi et al., 1998) responsible for lipid peroxidation of β -lipoprotein lipids, regulation of lipid and lipoprotein metabolism as well as stabilization of hepatocytes (Devi and Devaki, 1998) to neutralize the cytotoxic action of lipid peroxide rich lipoproteins during liver damage.

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